








Advanced Time Series Analysis and Forecasting

Chapter 7: Regime-switching models



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **BET**: Bucharest Exchange Trading (index)
- **BFGS**: Broyden–Fletcher–Goldfarb–Shanno (quasi-Newton optimiser)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CHP**: Carrasco–Hu–Ploberger (test, 2014)
- **CRPS**: Continuous Ranked Probability Score
- **DLW**: Diebold–Lee–Weinbach (TVTP model, 1994)
- **ECB**: European Central Bank
- **ECM**: Error Correction Model
- **EM**: Expectation–Maximisation (algorithm)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **ESI**: Economic Sentiment Indicator (European Commission)
- **FFBS**: Forward Filtering, Backward Sampling
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **GNP**: Gross National Product
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HICP**: Harmonised Index of Consumer Prices
- **HMP**: Haas–Mittnik–Paolella (MS-GARCH, 2004)
- **HQ**: Hannan–Quinn (information criterion)

Acronyms used in this material (2/2)

- **KS**: Kolmogorov–Smirnov (test)
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MCMC**: Markov Chain Monte Carlo
- **MFM**: Modelling Financial Markets
- **ML**: Machine Learning
- **MS**: Markov Switching
- **MS-GARCH**: Markov-Switching GARCH
- **MS-VAR**: Markov-Switching Vector AutoRegression
- **MSI**: Markov-Switching Intercept
- **MSI-AR**: Markov-Switching Intercept AutoRegression
- **MSIAH**: Markov-Switching Intercept, Autoregressive parameters and Heteroskedasticity
- **MSIH**: Markov-Switching Intercept and Heteroskedasticity
- **MSIH-AR**: Markov-Switching Intercept and Heteroskedasticity AutoRegression
- **MSM**: Markov-Switching Mean (Krolzig notation)
- **MSM-AR**: Markov-Switching Mean AutoRegression
- **NBER**: National Bureau of Economic Research
- **PIT**: Probability Integral Transform
- **QPS**: Quadratic Probability Score
- **RMSE**: Root Mean Squared Error
- **ROL**: Romanian old leu (ISO 4217 code, until 2005)
- **STAR**: Smooth Transition AutoRegression
- **SWARCH**: Switching ARCH (Hamilton–Susmel)
- **TAR**: Threshold AutoRegression
- **TSA**: Time Series Analysis
- **TVTP**: Time-Varying Transition Probabilities
- **US**: United States
- **VAR**: Vector AutoRegression
- **VAT**: Value Added Tax

Today's question and route

- **Question**: when the data-generating process itself changes (recession and expansion, calm and panic, a peg and a float), how do we infer the regime, estimate the model, test it and forecast with it?
  - the regime is never observed: everything rests on its probability given the data
- **Route** of the chapter
  - the Hamilton filter and the Kim smoother derived; EM and numerical maximum likelihood; identification and label switching
  - testing the number of regimes; time-varying transition probabilities; Markov-switching VAR; Markov-switching GARCH
  - Bayesian estimation by Gibbs sampling; regimes, long memory and breaks; forecasting and its evaluation
  - applications: US business cycles, Romanian inflation and GDP, S&P 500 and BET, EUR/RON
- We build on TSA, Chapter 10 (Hamilton's model with statsmodels), Chapter 2 (threshold models, breaks) and Chapter 6 (state space, Bayesian filtering); Seminar 7 comes before this lecture

Learning outcomes

- Derive the Hamilton filter, the likelihood and the Kim smoother, and write them in numpy with a numerically stable recursion
- Estimate Markov-switching regressions, autoregressions and VARs by EM and by numerical maximum likelihood, and diagnose local and degenerate maxima
- Explain why the likelihood-ratio test for the number of regimes is non-standard, and test by bootstrap and by information criteria
- Specify time-varying transition probabilities, regime-dependent impulse responses and MS-GARCH models without path dependence
- Estimate a switching model by Gibbs sampling, handle label switching, and evaluate regime forecasts with scores and probability scores

Reading, data and tools

- Backbone: Hamilton (1994), Ch. 22; Kim and Nelson (1999); Krolzig (1997); Frühwirth-Schnatter (2006)
  - Surveys: Hamilton (2016); Ang and Timmermann (2012)
- Python Quantlets of this chapter: Quantlets/Ch_07
  - Hamilton filter, Kim smoother, EM, expanded-state ML, TVTP, MS-VAR, MS-GARCH and the Gibbs sampler written in `numpy`
  - comparison: `statsmodels` `MarkovRegression` and `MarkovAutoregression`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
US real GNP growth, 1951Q2–1984Q4 (Hamilton's data); US industrial production and leading indicator, 1948–1991 (Filardo's data)	data sets distributed with statsmodels	replication
US real GDP, unemployment rate, NBER recession quarters	FRED (GDPC1, UNRATE, USRECQ)	dating, MS-VAR, forecasts
Romanian real GDP (quarterly) and HICP (monthly)	Eurostat (namq_10_gdp, prc_hicp_minr)	Gibbs, inflation regimes
S&P 500, BET: daily closes	EODHD (data/market)	MS-GARCH, bull/bear volatility regimes
EUR/RON	ECB reference rate to June 2005, BNR reference rate from July 2005	

- All sources are public and need no account or key

From Markov to Hamilton



Andrey Markov (1856–1922), around 1886

Photo: unknown author (1886); public domain; Wikimedia Commons

- 1906–1913: Markov chains, a dependence that needs only the present state
- 1970: Baum et al. (1970): maximum likelihood for hidden Markov models (the Baum–Welch algorithm, an EM before EM); speech recognition Rabiner (1989)
- 1973: Goldfeld and Quandt (1973): switching regressions in econometrics
- 1989–1990: Hamilton (1989) dates US recessions from GNP alone; Hamilton (1990) gives the EM algorithm; Kim (1994) the smoother
- 1993–2006: Bayesian estimation Albert and Chib (1993), Chib (1996), Frühwirth-Schnatter (2001); MS-VAR Krolzig (1997); MS-GARCH Gray (1996), Haas, Mittnik and Paolella (2004)

Known from TSA and new here

- Known (TSA, Chapter 10): $y_t = \mu_{S_t} + \varepsilon_t$, transition matrix, expected durations, ergodic probabilities, filtered and smoothed probabilities from `statsmodels`
  - US recessions from GDP, Romanian growth regimes, the effect of 2020
- New: every step derived and coded; estimation as an optimisation problem with traps; inference on the number of regimes
  - extensions: transitions that depend on covariates, systems (MS-VAR), volatility dynamics (MS-GARCH), Bayesian estimation
- Chapter 2 split the sample by an observed threshold (TAR, STAR) or a date (breaks); here the switch is a latent Markov chain

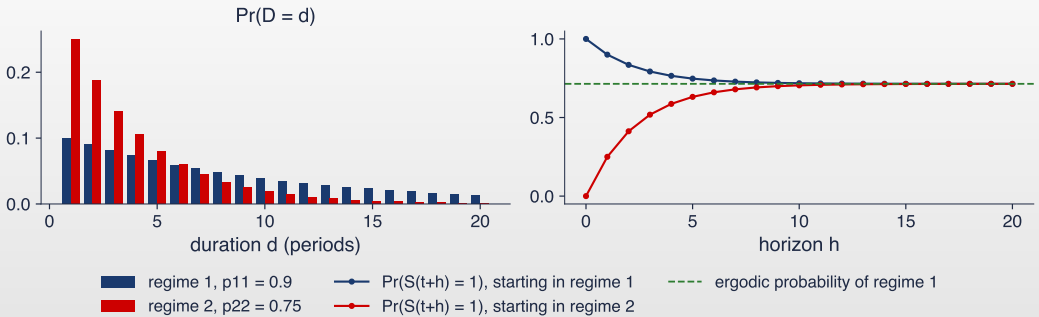
The model family (Krolzig's notation)

- MS(K)-AR(p): $y_t = \nu_{S_t} + \sum_{k=1}^{p} \phi_{k,S_t} y_{t-k} + \sigma_{S_t}\varepsilon_t$, $\varepsilon_t \sim$ i.i.d. $N(0,1)$, $S_t \in \{1,\dots,K\}$
  - S_t: the regime at t, one of K values; ν_{S_t}, ϕ_{k,S_t}, σ_{S_t}: the intercept, the AR(p) coefficients and the shock s.d. of the current regime
  - S_t a homogeneous Markov chain, $p_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$, independent of ε Krolzig (1997); $\mathrm{P} = (p_{ij})$: the transition matrix, rows sum to 1
- Which parameters switch gives the name:
  - **MSI**: intercept ν; **MSIH**: intercept and variance; **MSIAH**: intercept, AR coefficients and variance
  - **MSM** (Hamilton 1989): the *mean* μ_{S_t} switches, $y_t - \mu_{S_t} = \sum_k \phi_k (y_{t-k} - \mu_{S_{t-k}}) + \sigma\varepsilon_t$
- MSI: after a switch the mean adjusts gradually; MSM: it jumps at once; MSM needs the regimes of $p+1$ periods
- Special cases: $p_{ij} = \pi_j$ for all i is an i.i.d. mixture; an upper-triangular P is a change-point model Chib (1998)

The chain: ergodicity, durations, persistence

- Ergodic probabilities: $\boldsymbol{\pi}'\mathrm{P} = \boldsymbol{\pi}'$, $1'\boldsymbol{\pi} = 1$; with $K = 2$, $\pi_1 = (1 - p_{22})/(2 - p_{11} - p_{22})$
  - $\boldsymbol{\pi}$: the long-run share of time spent in each regime; 1: a vector of ones
  - durations are geometric: $\Pr(D_i = d) = p_{ii}^{d-1}(1 - p_{ii})$, $\mathbb{E}D_i = 1/(1 - p_{ii})$ (memoryless); D_i: the number of periods a visit to regime i lasts
- AR(1) representation Hamilton (1994), §22.2: $\xi_t = 1\{S_t = 1\}$ satisfies $\xi_t = (1 - p_{22}) + \lambda\xi_{t-1} + v_t$, $\lambda = p_{11} + p_{22} - 1$
  - $1\{\cdot\}$: the indicator (1 if true, 0 otherwise); v_t: the innovation; it is a martingale difference, not independent of the past (its variance depends on S_{t-1})
  - $\Pr(S_{t+h} = 1 \mid S_t) \to \pi_1$ at the rate λ^h: λ is the persistence of the regime process
- Duration dependence (recessions that are more likely to end the longer they last) needs an extended state or a semi-Markov chain Maheu and McCurdy (2000)

Durations and the speed of mixing



- $p_{11} = 0.9$, $p_{22} = 0.75$: expected durations 10 and 4 periods; $\pi_1 = 0.714$; $\lambda = 0.65$
- The gap to the ergodic probability shrinks by the factor λ each period: after 10 periods the starting regime is almost forgotten

ATS_ch7_hamilton

The Hamilton filter, derived

- Notation: $Y_t = (y_1, \ldots, y_t)$, $\hat{\xi}_{t|s}$ the K-vector $\Pr(S_t = j \mid Y_s)$, η_t the vector of densities $f(y_t \mid S_t = j, Y_{t-1})$
  - **prediction** (law of total probability, Markov property): $\hat{\xi}_{t|t-1} = \mathrm{P}' \hat{\xi}_{t-1|t-1}$
  - **update** (Bayes): $\hat{\xi}_{t|t} = \dfrac{\hat{\xi}_{t|t-1} \odot \eta_t}{1'(\hat{\xi}_{t|t-1} \odot \eta_t)}$
  - **likelihood**: $f(y_t \mid Y_{t-1}) = 1'(\hat{\xi}_{t|t-1} \odot \eta_t)$, so $\ln L = \sum_t \ln f(y_t \mid Y_{t-1})$
- Start: $\hat{\xi}_{1|0} = \boldsymbol{\pi}$ (ergodic) or a free parameter; $\odot$ is the element-wise product
- Same structure as the Kalman filter (Chapter 6): predict, update, prediction-error decomposition; here the state is discrete and the update is exact

Numerics and cost

- Densities underflow (e.g. 10^{-300} after a large outlier): work with $\ln \eta_t$ and subtract $m_t = \max_j \ln \eta_{jt}$
  - $\ln f(y_t \mid Y_{t-1}) = m_t + \ln \sum_j \hat{\xi}_{j,t|t-1} e^{\ln \eta_{jt} - m_t}$ (log-sum-exp); the filtered vector is unchanged
- Cost: $O(TK^2)$; with MSM-AR(p) the state is $(S_t, \ldots, S_{t-p})$, $M = K^{p+1}$ (32 for Hamilton's $K = 2$, $p = 4$)
  - the expanded transition matrix is sparse: from $(i_0, \ldots, i_p)$ only to $(j, i_0, \ldots, i_{p-1})$, with probability $p_{i_0 j}$
  - MSI-AR(p) needs only K states: the lagged y enter as regressors
- Our numpy filter and `statsmodels` give the same log-likelihood to machine precision when the initial distribution is the same

The Kim smoother, derived

- Goal: $\hat{\xi}_{t|T}$, the regime probabilities given the whole sample Kim (1994)
  - key step: $\Pr(S_t = i \mid S_{t+1} = j, Y_T) = \Pr(S_t = i \mid S_{t+1} = j, Y_t)$: given S_{t+1}, future y carry no extra information on S_t
  - exact for MSI/MSIH/MSM on the expanded state; an approximation only when the state also has a continuous part (Kim 1994, Chapter 6)
  - proof of the key step: Appendix
- Bayes on the pair: $\Pr(S_t = i, S_{t+1} = j \mid Y_T) = \dfrac{\hat{\xi}_{i,t|t}\, p_{ij}}{\hat{\xi}_{j,t+1|t}}\, \hat{\xi}_{j,t+1|T}$
  - sum over j: $\hat{\xi}_{t|T} = \hat{\xi}_{t|t} \odot \left[\mathrm{P}(\hat{\xi}_{t+1|T} \oslash \hat{\xi}_{t+1|t})\right]$, backwards from $\hat{\xi}_{T|T}$; $\oslash$: element-wise division
- The joint probabilities of (S_t, S_{t+1}) are the E-step of EM; sampling instead of summing gives FFBS (section Bayesian estimation)

Predicted, filtered and smoothed regime probabilities

- **Predicted** $\hat{\xi}_{t+1|t}$: what a forecaster expects for next period; it enters the forecast density
- **Filtered** $\hat{\xi}_{t|t}$: what could be known at t; the real-time signal (Chauvet and Piger 2008)
  - evaluate real-time dating with filtered probabilities and data vintages, never with smoothed ones
- **Smoothed** $\hat{\xi}_{t|T}$: the historian's answer; it uses the future and is revised as data arrive
  - the dating of past regimes, and the weights of the M-step
- Confusing them is the most common error in applied work: a smoothed probability is not a forecast

EM: the complete-data likelihood

- If $S_1, \ldots, S_T$ were observed, the log-likelihood would split Dempster, Laird and Rubin (1977), Hamilton (1990):
  - $\ln L_c = \sum_j 1\{S_1 = j\} \ln \rho_j + \sum_{t \geq 2} \sum_{i,j} 1\{S_{t-1} = i, S_t = j\} \ln p_{ij} + \sum_t \sum_j 1\{S_t = j\} \ln \eta_{jt}(\theta_j)$
  - $\rho_j = \Pr(S_1 = j)$: initial probabilities; $\eta_{jt}(\theta_j)$: the density of y_t in regime j, with parameters θ_j
  - transition counts and regime-by-regime regressions: closed forms
- **E-step**: replace the indicators by their expectations given Y_T and the current $\theta^{(m)}$ (m: the iteration)
  - $\hat{\xi}_{j,t|T}$ and $\hat{\xi}_{ij,t|T} = \Pr(S_{t-1} = i, S_t = j \mid Y_T)$: exactly the output of the Kim smoother
- Baum et al. (1970) derived the same recursion for hidden Markov models (the forward–backward algorithm)

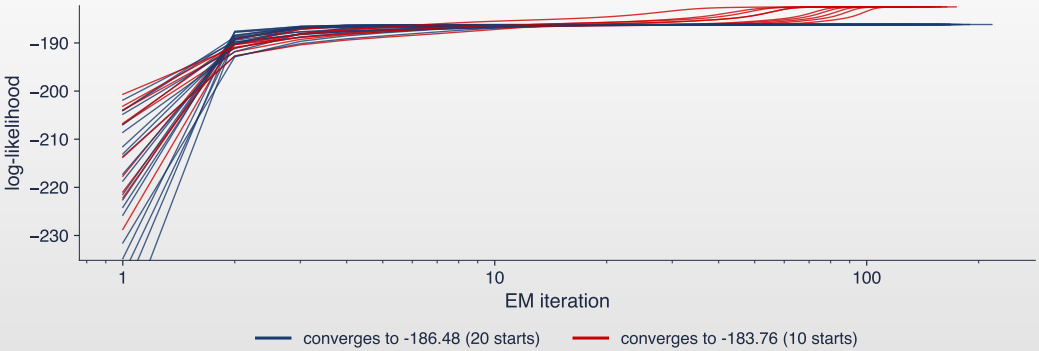
EM: the M-step in closed form

- Transitions: $p_{ij}^{(m+1)} = \dfrac{\sum_{t=2}^{T} \hat{\xi}_{ij,t|T}}{\sum_{t=2}^{T} \hat{\xi}_{i,t-1|T}}$ (expected transitions over expected visits); $\rho^{(m+1)} = \hat{\xi}_{1|T}$
- Switching coefficients (MSIAH): weighted least squares $\beta_j^{(m+1)} = (X' W_j X)^{-1} X' W_j y$, $W_j = \mathrm{diag}(\hat{\xi}_{j,t|T})$; X: rows x_t' (constant and lags), y: the observations; each date weighs by its probability of being in regime j
- Variances: $\sigma_j^{2(m+1)} = \sum_t \hat{\xi}_{j,t|T} (y_t - x_t'\beta_j)^2 / \sum_t \hat{\xi}_{j,t|T}$; MS-VAR: the same with matrices
- Common coefficients (MSIH-AR): one stacked weighted regression with weights $\hat{\xi}_{j,t|T}/\sigma_j^2$ (a conditional M-step, ECM)
- MSM-AR: μ and ϕ enter as products, so no closed form: numerical maximisation (or EM with a numerical M-step)
- Derivation of the M-step for P and of the monotonicity of EM: ▸ Appendix

EM and numerical ML in practice

- EM never lowers the likelihood, is robust far from the optimum, but slow near it (linear convergence)
  - monotonicity needs ρ as a free parameter; with $\rho = \boldsymbol{\pi}(\mathrm{P})$ the M-step for P is only approximate
- Numerical ML: BFGS on unconstrained parameters ($\ln \sigma_j^2$, logits of p_{ij}); standard errors from the Hessian by the delta method
  - practice: EM from many starts, then a Newton-type polish and the Hessian
- The likelihood is unbounded: $\sigma_j \to 0$ around one observation with $p_{jj} \to 0$ gives $\ln L \to \infty$
  - remedies: a lower bound on σ_j / σ_k Hathaway (1985), priors (section Bayesian estimation), or reject regimes that last one period
- Standard errors near the boundary ($p_{ii} \approx 1$) are unreliable: report profile likelihoods or bootstrap intervals

EM from 30 starting values



- MSIH(2)-AR(1) on Hamilton's GNP data: switching intercept and variance, common AR(1) coefficient; random starting values; log scale for iterations

ATS_ch7_estimation

Interpreting the EM paths

- 2 limits (log-likelihood, number of starts): -186.48 (20); -183.76 (10); median 159 iterations, at most 216
- The highest (-183.76) is **degenerate**: a regime with $\sigma^2 = 0.049$, $p_{11} < 10^{-9}$ and mean 1.11, used by 29 isolated quarters
- `statsmodels` (20 random searches) reports -185.73, a third local maximum that none of our starts reached
- Lesson: report the starting-value design, all local maxima, and why the chosen one is economically meaningful

Identification and label switching

- The likelihood is invariant to the $K!$ permutations of the labels: $(\theta_1, \theta_2, \mathrm{P})$ and $(\theta_2, \theta_1, \mathrm{P}'')$ fit equally well
  - for ML this is harmless: pick one mode and name the regimes by an ordering ($\mu_1 < \mu_2$ or $\sigma_1 < \sigma_2$)
  - for MCMC it is not: the sampler jumps between modes and posterior means of μ_1 average the regimes Celeux, Hurn and Robert (2000), Stephens (2000)
- Identification also fails locally: K regimes with $\theta_i = \theta_j$, or a regime never visited, leave P unidentified
  - this is the root of the testing problem of the next section
- Statistical regimes need not be economic regimes: the same GDP data give recessions (MSM), volatility eras (MSIH) or single outliers

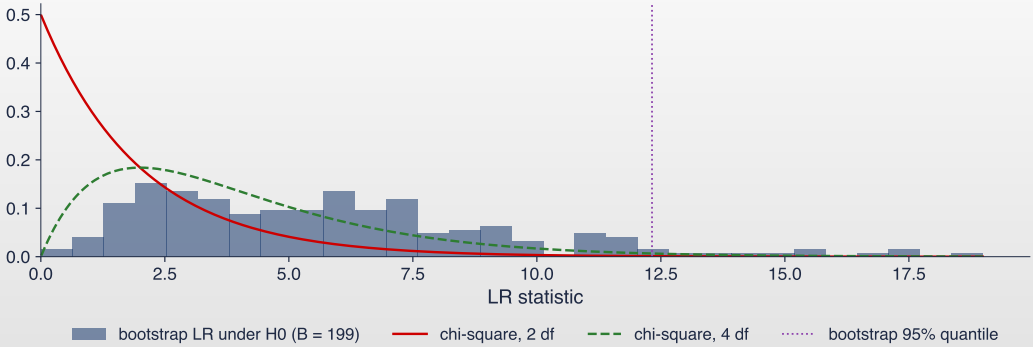
The non-standard LR test

- H_0: one regime ($\mu_1 = \mu_2$, $\sigma_1 = \sigma_2$) against H_1: two regimes
  - (i) under H_0, p_{11} and p_{22} are **not identified**: nuisance parameters present only under H_1, the Davies problem of Chapter 2 Davies (1987)
  - (ii) the null can be written as $p_{11} = 1$: a parameter on the **boundary**
  - (iii) the score with respect to $\mu_2 - \mu_1$ is **identically zero** at H_0: the information matrix is singular
- Each of the three breaks one of the conditions for $2 \ln \Lambda \to \chi^2_q$ (Λ: the likelihood ratio; q: the number of restrictions): counting parameters does not give the critical value
- Same structure as testing linearity against TAR or STAR (Chapter 2), with a latent instead of an observed switch

Solutions in the literature

- Hansen (1992): treat the likelihood as an empirical process in the nuisance parameters and bound the standardised LR; conservative, heavy to compute
  - applied to Hamilton's GNP model: no evidence against one regime at conventional levels
- Garcia (1998): the asymptotic null distribution of the sup-LR for MS models with switching mean (and variance), with tabulated critical values
- Cho and White (2007): quasi-LR against a two-component mixture (an i.i.d. switch has the same null behaviour)
- Carrasco, Hu and Ploberger (2014): an optimal score-type test against Markov switching that needs only the null model
- Practice: **parametric bootstrap** of the LR
  - simulate from the estimated null; re-estimate both models (with several starts) on each sample
  - $p = (1 + \#\{LR^* \geq LR\})/(B + 1)$; LR^*: the B bootstrap statistics; $\#$: their count above LR
- Selection rather than testing: AIC, BIC, HQ, or Bayes factors through the marginal likelihood Chib (1995)

The bootstrap null distribution of the LR statistic



- US GDP growth, 1947Q2–2019Q4 ($T = 290$): H_0 Gaussian AR(1), H_1 MSIH(2)-AR(1); 199 samples simulated from the estimated AR(1), both models re-estimated, EM with 4 starts

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Interpreting the test

- Bootstrap 95% quantile 12.32, against 5.99 (χ^2_2) and 9.49 (χ^2_4); mean of the null draws 5.79: counting parameters gets the critical value wrong
- Observed LR = 71.7, bootstrap p-value 0.005 (the smallest possible with $B = 199$): two regimes
- But which regimes? Standard deviations 0.46 and 1.11 pp, $p_{ii} = 0.990$ and 0.985: volatility eras (the Great Moderation), not recessions
- BIC: 755.5 ($K = 1$), 706.4 ($K = 2$), 734.7 ($K = 3$); AIC: 744.5, 680.8, 687.0

The paper



Chicago, February 1931: the regime every forecaster fears

Photo: US National Archives (1931); public domain; Wikimedia Commons

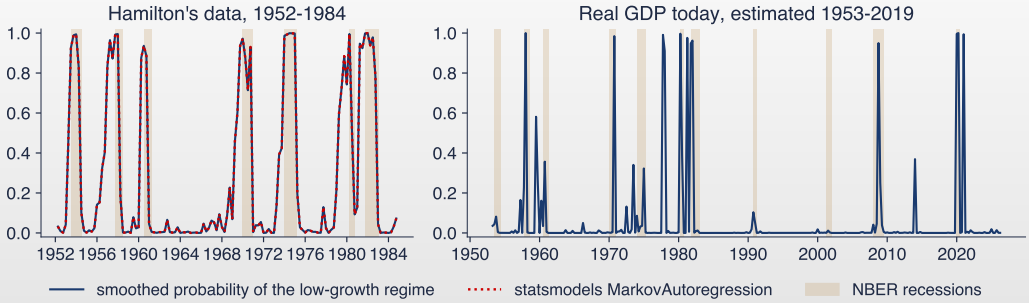
- Hamilton (1989), Section 4: US real GNP, 100 × log change, 1951Q2–1984Q4, MSM(2)-AR(4)
- Equation (4.3): $y_t - \mu_{S_t} = \sum_{k=1}^{4} \phi_k (y_{t-k} - \mu_{S_{t-k}}) + \sigma \varepsilon_t$; estimates in Table I
- The claim: the dates where the low-growth regime is likely match the NBER recessions, although the NBER dates were never used
- We replicate on his data set, compare numpy with `statsmodels`, then re-estimate on today's GDP

Replication: Hamilton's data, two implementations

Estimate	μ_0	μ_1	ϕ_1	ϕ_2	ϕ_3	ϕ_4	σ	$\ln L$
numpy	-0.359	1.164	0.013	-0.058	-0.247	-0.213	0.769	-181.26
(standard error)	(0.265)	(0.075)	(0.120)	(0.138)	(0.107)	(0.111)		
statsmodels	1.164	-0.359	0.013	-0.058	-0.247	-0.213	0.769	-181.26

- Transitions: $p_{00} = 0.755$ (stay in the low regime), $p_{11} = 0.904$; expected durations 4.1 and 10.4 quarters; $T = 131$ after 4 lags
- Largest difference between the two smoothed probability paths: $< 10^{-4}$
- Estimation: BFGS on the expanded state of 32 regime combinations, 8 random starts; standard errors from the numerical Hessian

Hamilton's regimes then and now



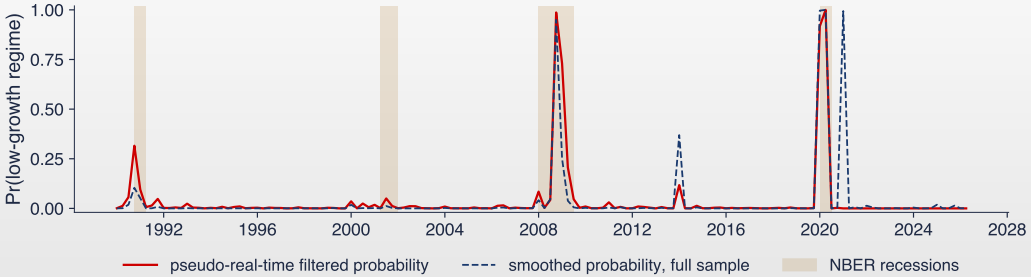
⊡ Left: smoothed probability of the low-growth regime on Hamilton's data. Right: the same specification on today's GDPC1, estimated on 1953Q2–2019Q4 ($T = 267$), probabilities to 2026Q2 with the same parameters

ATS_ch7_hamilton

Interpreting the replication

- On his data, the regime tracks the NBER: QPS 0.179 and 89% concordance (probability above 0.5 against the NBER quarters) Diebold and Rudebusch (1989), Harding and Pagan (2002)
- On today's data the low regime changes nature: mean -1.22%, $p_{00} = 0.26$ (duration 1.4 quarters): single sharp falls, not recessions
- It catches 2008Q4 (0.95) and 2020Q2 (1.00) but not 2001 (at most 0.01); QPS 0.229 on 37 NBER quarters
- After 1984 recessions are rarer and milder (the Great Moderation): a fixed two-mean model loses its grip; this motivates TVTP, MSIH and three regimes

Dating in pseudo real time



- Hamilton's model re-estimated every year on the data available then (current vintage, no data revisions); filtered probability of each quarter with information up to that quarter, 1990–2026

ATS_ch7_hamilton

Interpreting real-time dating

- First quarter above 0.5: 1990–91 none (maximum 0.32), 2001 none (maximum 0.05), 2008 2008Q4, 2020 2020Q1
- QPS before 2020: 0.127 in real time against 0.152 smoothed; 0 false signals above 0.5
- Chauvet and Piger (2008): with real-time vintages, MS models call the start of recessions faster than the NBER announcement, but mild recessions are missed
- Our exercise flatters the model: today's revised GDP was not available in 2001

Letting the chain depend on observables

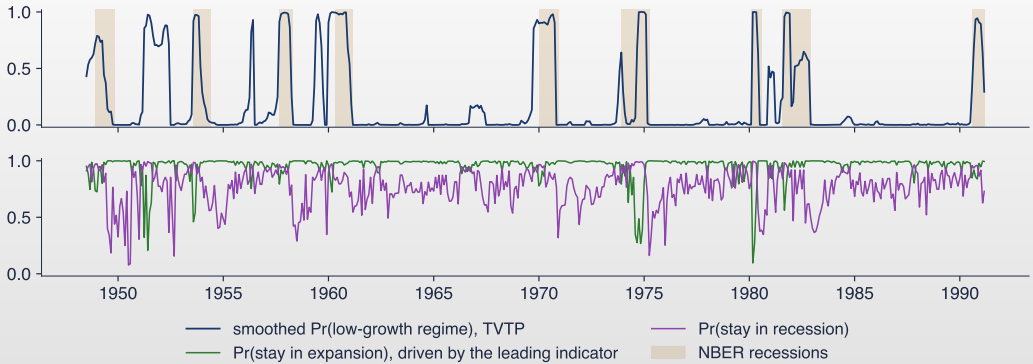
- Diebold, Lee and Weinbach (1994), Filardo (1994): $p_{ii,t} = \Pr(S_t = i \mid S_{t-1} = i, z_{t-1}) = \dfrac{\exp(z'_{t-1}\gamma_i)}{1 + \exp(z'_{t-1}\gamma_i)}$
  - z_{t-1}: observed variables known at $t-1$ (with a constant); γ_i: their coefficients; the logistic function keeps $p_{ii,t}$ in $(0,1)$
  - a leading indicator, the term spread or a policy rate can make a recession more or less likely to begin or end
  - expected durations now vary over time; the chain is no longer homogeneous
- Filter and smoother unchanged, with P_t in place of P; EM: the M-step for γ_i is a weighted logit (DLW)
  - z_{t-1} must be predetermined and must not depend on S_t; if z reacts to the regime, the model is misspecified (endogenous switching)
- Test of constant transitions: H_0: slopes of $\gamma_i = 0$ is a standard LR test, since the regimes exist under both hypotheses

Replication: Filardo (1994)

Model	μ_0	μ_1	γ_0	γ_1	ln L	statsmodels
TVTP, numpy	-0.87	0.50	(1.49; -1.09)	(4.33; 1.75)	-586.20	-586.20
(standard error)			(0.48; 0.62)	(0.76; 0.52)		
constant transitions		2.77	$p_{00} = 0.983$	$p_{11} = 0.14$	-590.67	

- US industrial production growth, monthly, $T = 513$ (Filardo's data); MSM(2)-AR(4); z_{t-1} = (1, growth of the composite leading indicator)
- LR of TVTP against constant transitions: 8.94 ($p = 0.011$, 2 restrictions)
- The constant-transition maximum is not a business cycle: a high-mean regime (2.77% per month) that lasts about one month

Transition probabilities driven by the leading indicator



- Top: smoothed probability of the low-growth regime; bottom: the probabilities of staying in each regime, month by month

ATS_ch7_tvtp

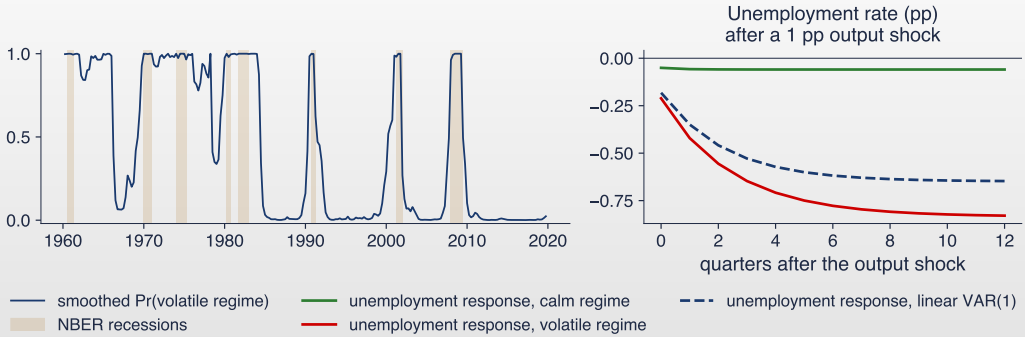
Interpreting TVTP

- Stay in expansion: median 0.988, but down to 0.10 when the leading indicator falls sharply: expansions end when the indicator turns
- Stay in recession falls when the indicator rises (γ slope -1.09): recoveries are announced by the indicator
- QPS against the NBER months 0.197, concordance 88%
- Caveat: the leading indicator was built knowing the NBER dates; a real-time test needs its vintages

MS-VAR and regime-dependent responses

- MSIAH(K)-VAR(p): $y_t = \nu_{S_t} + \sum_{k=1}^{p} A_{k,S_t} y_{t-k} + \Sigma_{S_t}^{1/2} \varepsilon_t$, $y_t \in \mathbb{R}^n$ Krolzig (1997)
  - ν_{S_t}, A_{k,S_t}, Σ_{S_t}: intercept vector, lag matrices and error covariance of the current regime; $\varepsilon_t \sim N(0, I_n)$
  - EM: the M-step is multivariate weighted least squares by regime; $\Sigma_j = \sum_t \hat{\xi}_{j,t|T} \hat{u}_{jt} \hat{u}_{jt}' / \sum_t \hat{\xi}_{j,t|T}$ ($\hat{u}_{jt}$: residuals of regime j)
  - parameters grow as $K(n + n^2 p + n(n+1)/2)$: shrinkage or restricted switching (MSIH) in larger systems (Chapter 5)
- **Regime-dependent impulse responses** Ehrmann, Ellison and Valla (2003): $\partial y_{t+h} / \partial \varepsilon_t$ computed with (A_j, Σ_j), assuming regime j persists over the horizon
  - the full response averages over future regime paths and depends on the current regime probabilities; simulate it
- Switching in Σ alone identifies structural shocks by heteroskedasticity (Chapter 3); Sims and Zha (2006) ask whether US monetary policy switched, and find mainly variance switches

A two-regime VAR for output and unemployment



- MSIAH(2)-VAR(1) for US GDP growth and the change of the unemployment rate, 1960Q1–2019Q4 ($T = 239$, 20 parameters), EM from 12 starts; Cholesky order: output first; responses scaled to a 1 pp output shock

ATS_ch7_msvar

Interpreting the MS-VAR

- The regimes are calm and volatile
  - output shock s.d. 0.42 against 1.02 pp; durations 21.8 and 14.3 quarters
  - the volatile regime holds 83% of quarters before 1984, 17% after, and 100% of NBER quarters
- Unemployment 12 quarters after a 1 pp output shock: -0.06 pp (calm), -0.83 pp (volatile), -0.65 pp (linear VAR)
- Okun's law is regime-dependent
  - in the calm regime output surprises barely move unemployment (correlation of shocks -0.15, against -0.69)
  - a linear VAR averages two transmissions
- In L: -166.8 (MS-VAR) against -222.0 (linear VAR); regime-dependent responses assume the regime persists over the horizon

Regimes in volatility



Lehman Brothers, Times Square, 2007: a calm regime about to end

Photo: David Shankbone (2007); CC BY-SA 3.0; Wikimedia Commons

- Lamoureux and Lastrapes (1990): structural shifts in the variance bias GARCH persistence $\alpha + \beta$ towards 1
- Hamilton and Susmel (1994): SWARCH, an ARCH whose scale switches with a Markov chain; most of the persistence moves into the chain
- MSIH on returns: the simplest volatility-regime model, already a useful benchmark for GARCH Bollerslev (1986)
- Chapter 8 develops realised measures and multivariate GARCH; here: what regimes do to GARCH

The path-dependence problem

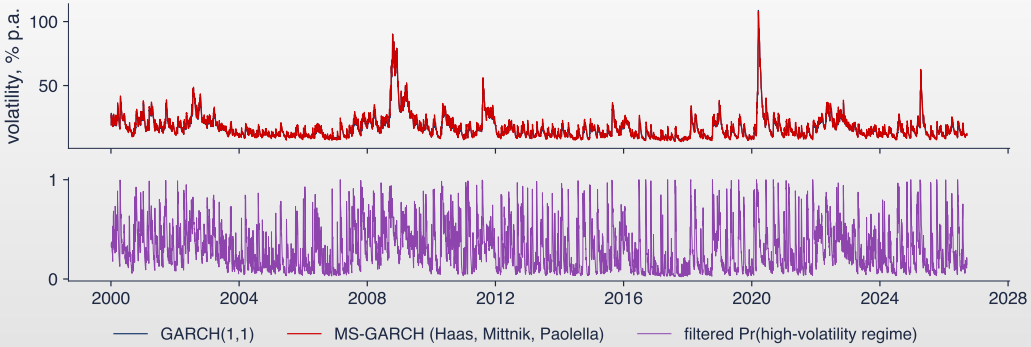
- Naive MS-GARCH: $h_t = \omega_{S_t} + \alpha_{S_t}\varepsilon_{t-1}^2 + \beta_{S_t} h_{t-1}$
  - h_t: the conditional variance; ε_{t-1}: the last return shock; ω, α, β: the GARCH parameters of the regime
  - h_{t-1} is itself regime-dependent
  - h_t depends on the whole path $(S_1, \ldots, S_t)$: K^t histories, so the Hamilton filter cannot be applied ($2^{100} \approx 10^{30}$)
- Gray (1996): collapse the past into $h_{t-1} = \mathbb{E}[\varepsilon_{t-1}^2 \mid Y_{t-2}]$, the variance of the mixture: $h_{j,t} = \omega_j + \alpha_j \varepsilon_{t-1}^2 + \beta_j h_{t-1}$
  - Klaassen (2002): condition on S_t as well, using $\Pr(S_{t-1} \mid S_t, Y_{t-1})$
- Haas, Mittnik and Paolella (2004): K GARCH processes run in parallel, $h_{j,t} = \omega_j + \alpha_j \varepsilon_{t-1}^2 + \beta_j h_{j,t-1}$; the chain picks one
  - no path dependence, exact likelihood, explicit stationarity conditions; asymptotics in Bauwens, Preminger and Rombouts (2010)

S&P 500: GARCH against MS-GARCH

Model	$\alpha+\beta$, regime 1	$\alpha+\beta$, regime 2	p_{ii}	$\ln L$	BIC
GARCH(1,1)	0.981			−9219.9	18475.0
MS-GARCH (HMP)	0.964	0.998	0.959; 0.925	−9119.1	18317.5
MS-GARCH (Gray)	1.000	0.921	0.981; 0.956	−9207.4	18494.1

- Daily log returns (%), January 2000 – September 2026, $T = 6\,714$; Normal innovations, common mean; numerical ML in numpy from several starting values
- HMP regimes: average conditional volatility while in the regime 13.1% and 27.6% per year; expected durations 24 and 13 days

Conditional volatility and the high-volatility regime



- Top: annualised conditional volatility, GARCH(1,1) and MS-GARCH (HMP, mixture variance); bottom: filtered probability of the high-volatility regime

ATS_ch7_msgarch

Interpreting MS-GARCH

- BIC prefers HMP (18317.5) to GARCH (18475.0); the high-volatility regime is active on 21% of days
- Single-regime persistence 0.981; within the regimes 0.964 (calm) and 0.998 (high volatility): regimes do not remove persistence automatically, they move it to where it belongs
- Gray's collapsing fits worse here (-9207.4): the approximation is not innocuous
- Local maxima are frequent: different starts give different regime splits (always report the starting design)

Equity regimes: S&P 500 and BET

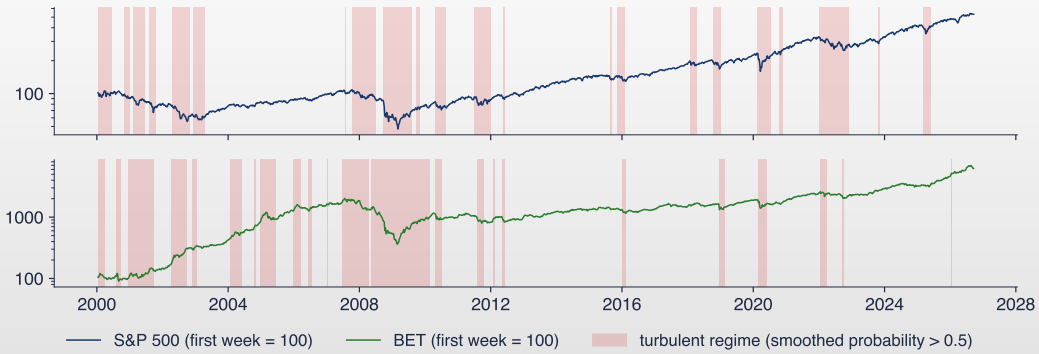


Bucharest Stock Exchange, 2024

Photo: Corina Chitu (2024); CC BY-SA 4.0; Wikimedia Commons

- Weekly log returns, 2000–2026, MSIH(2): $r_t = \mu_{S_t} + \sigma_{S_t}\varepsilon_t$ Maheu and McCurdy (2000), Ang and Timmermann (2012)
- S&P 500 (annualised): calm 16.6% mean, 10.9% volatility; turbulent −20.6%, 28.3%
- BET: calm 20.1%, 13.6%; turbulent 3.2%, 39.9%
- Durations (weeks), calm and turbulent: S&P 500 37 and 14; BET 33 and 12

Turbulent regimes on two markets



- Log index levels (first week = 100); shaded: smoothed probability of the turbulent regime above 0.5; EM from 15 starts and a numerical polish (`statsmodels`: same $\ln L$)

ATS_ch7_bullbear

Interpreting the equity regimes

- Turbulent share: S&P 500 26%, BET 25%; correlation of the two turbulent probabilities 0.48; both turbulent in 15% of weeks
- The regimes are mostly volatility regimes: the turbulent mean is lower but estimated with large error
- $\ln L$: S&P 500 -3019.3 (numpy) and -3019.3 (`statsmodels`); BET -3339.9 and -3339.9
- The BET has its own turbulent episodes (local politics, the 2007–2009 crash), not only imported ones

Regimes and asset allocation, briefly

- Ang and Bekaert (2002)
  - with regime switching, optimal portfolios depend on the current regime probabilities
  - home bias and the value of international diversification change across regimes
- Mean-variance weight of equity against cash, risk aversion $\gamma = 5$: $w = \mu/(\gamma\sigma^2)$; μ, σ^2: weekly mean and variance of the return in the regime
  - S&P 500: 2.79 in the calm regime, -0.51 in the turbulent one, 0.39 with constant moments
  - with one-step mixture moments the weight moves between -0.47 and 2.02
- Caveats: estimation error in regime means is large; out-of-sample gains must be tested (Chapter 1) and transaction costs included
- Market-risk applications (VaR and ES with regimes) are in MFM, Chapter 7

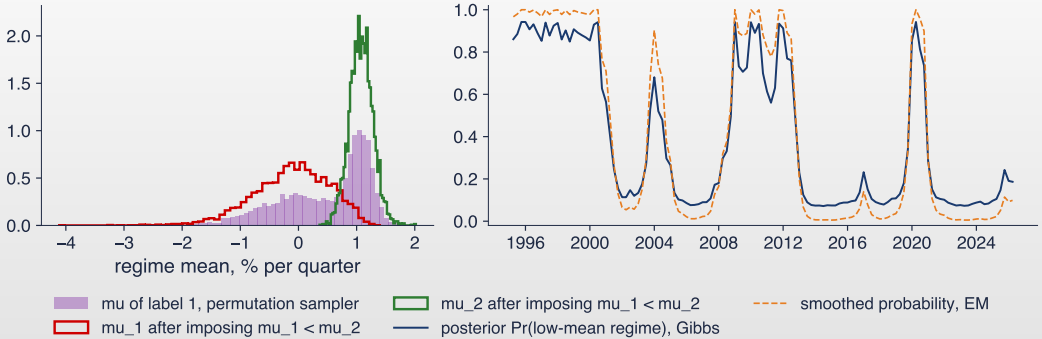
Gibbs sampling with data augmentation

- Treat $S = (S_1, \ldots, S_T)$ as parameters Albert and Chib (1993): given S, the model is a set of regressions; given θ, S is a hidden Markov chain
  - 1. $S \mid \theta, Y$; 2. P $\mid S$: rows $\sim$ Dirichlet$(a_{i\cdot} + n_{i\cdot})$, n_{ij} the transition counts, a_{ij} the prior Dirichlet parameters
  - 3. $\mu_j \mid \sigma_j^2, S, Y \sim$ Normal and $\sigma_j^2 \mid \mu_j, S, Y \sim$ inverse gamma: conjugate updates on the observations of regime j (Chapter 5)
- **FFBS** Chib (1996): draw the whole path at once: $S_T \sim \hat{\xi}_{T|T}$, then $\Pr(S_t = i \mid S_{t+1} = j, Y_t) \propto \hat{\xi}_{i,t|t} p_{ij}$ backwards
  - single-move sampling of S_t given S_{t-1}, S_{t+1} mixes very slowly when regimes are persistent
  - the discrete analogue of the simulation smoother of Chapter 6 Carter and Kohn (1994)
- Priors keep σ_j away from 0 and p_{ii} away from the boundary: the degenerate maxima of ML disappear

Label switching in MCMC, and choosing K

- **Random permutation sampler** Frühwirth-Schnatter (2001): permute the labels at random after each sweep; the sampler then visits all $K!$ modes
  - the unconstrained output shows which parameters separate the regimes; identify afterwards by the constraint that does
  - a bad constraint (e.g. on parameters that barely differ) cuts a mode in half and biases both regimes Celeux, Hurn and Robert (2000)
- Choosing K: marginal likelihood $p(Y \mid K)$ from the Gibbs output Chib (1995), $\ln p(Y) = \ln f(Y \mid \theta^*) + \ln p(\theta^*) - \ln p(\theta^* \mid Y)$
  - the likelihood ordinate comes from the Hamilton filter; the posterior ordinate from reduced Gibbs runs
- Change-point models (Chib 1998) are the same sampler with an upper-triangular P

Romanian GDP growth: two regimes by Gibbs sampling



- Quarterly growth, 1995Q2–2026Q2 ($T = 125$), MSIH(2) without AR term; 5 000 draws after burn-in; left: permutation sampler, right: sampler identified by $\mu_1 < \mu_2$

ATS_ch7_bayes

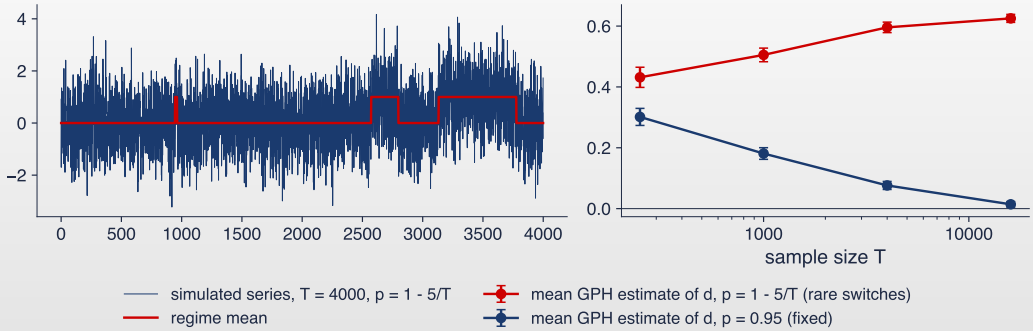
Interpreting the posterior

- Permutation sampler: label 1 carries the larger mean in 49% of draws: the bimodal histogram is the two regimes, not two answers
- Identified: μ_1 −0.11 [−1.26; 0.87], σ_1 3.80; μ_2 1.08 [0.76; 1.41], σ_2 1.57 (90% credible intervals)
- p_{11} 0.85 [0.73; 0.94], p_{22} 0.91; EM: $\mu = 0.03$ and 1.04, $p_{ii} = 0.90$ and 0.94
- The means overlap, the standard deviations do not: the regimes are a volatile era (1995–2000, 2009–2012, 2020) and a stable one; the variance is the better identifying constraint
- Posterior and EM regime probabilities correlate at 0.99; the posterior also carries the parameter uncertainty that the EM path ignores

Rare switches look like long memory

- Diebold and Inoue (2001): $y_t = \mu_{S_t} + \varepsilon_t$ with $p_{ii} = 1 - c/T^{\delta}$
  - $c, \delta > 0$: switches become rarer as T grows
  - the variance of partial sums grows like that of an I(d) process, $d > 0$
  - the autocorrelations $\propto \lambda^k$ with $\lambda \to 1$ decay so slowly that, in any finite sample, they look hyperbolic
- Consequences
  - a significant GPH or local Whittle estimate (Chapter 10) does not discriminate fractional integration from occasional breaks or regimes
  - the same holds for GARCH persistence (section Volatility regimes and MS-GARCH) and for unit-root tests with breaks (Chapter 2)
- Which description is "true" matters less than which one forecasts better out of sample

Simulation: GPH estimates of d under regime switching



- $y_t = \mu_{S_t} + \varepsilon_t$, $\mu \in \{0, 1\}$, $\sigma = 1$; mean GPH estimate ($m = T^{0.5}$) over 200 replications, ± 2 standard errors

ATS_ch7_longmem

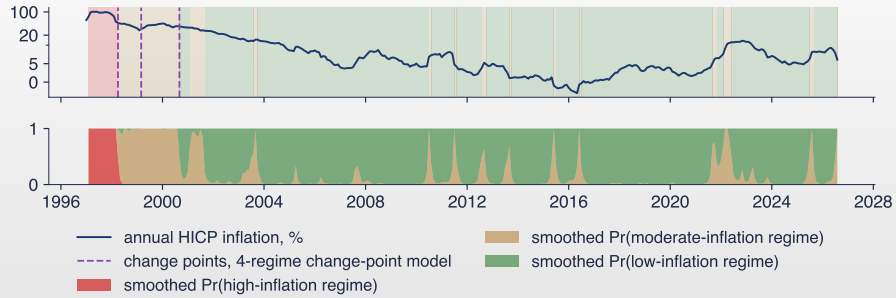
Interpreting the simulation

- Rare switches ($p = 1 - 5/T$): mean $\hat{d} = 0.43, 0.51, 0.60, 0.63$ for $T = 250, 1\,000, 4\,000, 16\,000$: it does not vanish as T grows
- Fixed $p = 0.95$: 0.30, 0.18, 0.08, 0.01: short memory, $\hat{d} \to 0$ as the frequencies used approach zero
- The path on the left has 6 switches in 4000 observations; its autocorrelation is 0.12 at lag 50 and 0.06 at lag 200

Regimes or breaks?

- A break (Chapter 2) is a regime that never returns: P upper bidiagonal, $S_1 = 1$ Chib (1998)
  - the Hamilton filter and EM apply unchanged; the transition mask keeps the forbidden transitions at zero
- Recurrent regimes pool information across episodes: the second high-inflation episode reuses the parameters of the first
  - breaks need a new parameter set for every episode, and say nothing about the future
- Compare the two by BIC or marginal likelihood; Garcia and Perron (1996): the US real interest rate is better described by three recurrent regimes than by a stable process

Romanian inflation regimes since 1997



Annual HICP inflation, $100 \ln(P_t / P_{t-12})$, from February 1997 ($T = 355$); MSIH(3)-AR(1), EM from 25 starts; dashed: change points of the 4-regime change-point model; symmetric log scale

ATS_ch7_ro_inflation

Interpreting the inflation regimes

- Regime levels $\nu_j/(1-\phi)$: 68.7%, 17.8% and 2.9%; shock s.d. 10.58, 1.70 and 0.45 pp; $\phi = 0.971$: after 2001 the two lower regimes differ mainly in volatility
- The volatile regime returns in the 2010 VAT increase, in 2022 and in 2025; filtered probabilities in August 2026 (inflation 6.1%): 0.00, 0.93, 0.07
- BIC prefers change points: 3 segments 887.0, 4 segments 897.5, Markov switching 934.1; the change points of the 4-segment model: April 1998, March 1999, September 2000
- Reading: disinflation was a one-way process
  - after 2000 a single, very persistent segment absorbs the 2022 and 2025 surges as shocks
  - whether they are a recurrent regime is decided out of sample

Romania's exchange-rate regimes

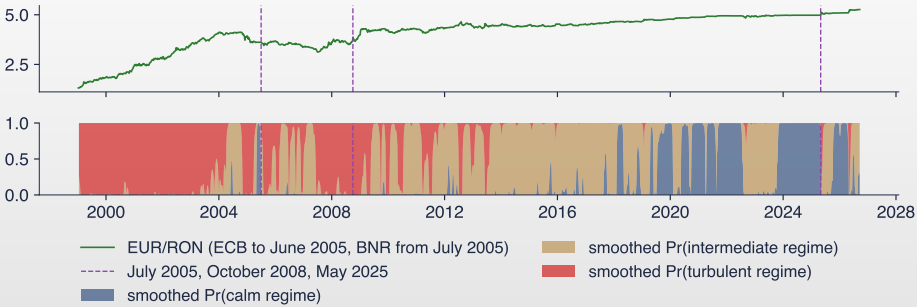


National Bank of Romania, Bucharest, 2014

Photo: Crislia (2014); CC BY-SA 4.0; Wikimedia Commons

- 1999–2005: crawling depreciation of the old leu under a managed float
- July 2005: redenomination (10 000 ROL = 1 RON); August 2005: inflation targeting with a managed float; capital account fully liberalised in 2006
- Episodes: October 2008, 2011–2012, March 2020, May 2025 (presidential election and fiscal stress)
- Model: weekly log changes, MSIH(3) ordered by volatility; regimes are policy outcomes, not the announced regime

EUR/RON volatility regimes, 1999–2026



- Top: EUR/RON (ECB reference rate to June 2005, BNR from July 2005); bottom: smoothed probabilities of the calm, intermediate and turbulent regimes; $T = 1\,445$ weeks

ATS_ch7_eurron

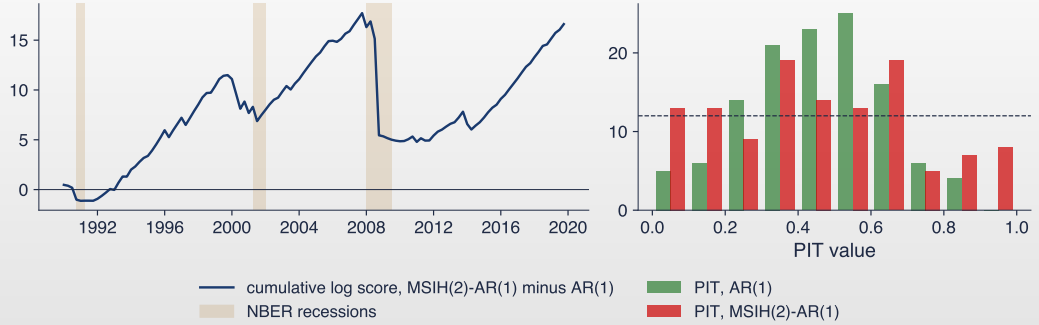
Interpreting the EUR/RON regimes

- Weekly s.d. by regime: 0.07, 0.39 and 1.51%; expected durations 16, 17 and 20 weeks
- Before July 2005 the turbulent regime holds 87% of weeks (the steady depreciation of the old leu, mean 0.25% per week), after it 18%; the calm regime goes from 2% to 25%
- Turbulent probability peaks at 1.00 in October 2008–March 2009 and at 1.00 in spring 2025; the largest 2025 weekly change, 2.76%, in the week ending 9 May 2025
- EUR/RON averaged 4.9773 in April 2025 and stood at 5.2644 at the end of the sample: a level shift followed by a return to calm, as a managed float predicts

Forecasts from a switching model

- Regime forecast: $\hat{\xi}_{T+h|T} = (\mathrm{P}')^h \hat{\xi}_{T|T}$, converging to $\boldsymbol{\pi}$ at the rate λ^h
  - one step: $f(y_{T+1} \mid Y_T) = \sum_j \hat{\xi}_{j,T+1|T}\, N(x'_{T+1}\beta_j, \sigma_j^2)$, a **mixture**: skewed, fat-tailed, possibly bimodal
- Several steps with AR terms: the mean is not linear in past regimes (MSM) or needs the regime path; simulate paths of (S, y)
- Evaluation (Chapter 1): point forecasts by RMSE and Diebold–Mariano; densities by log score, PIT and CRPS; regime probabilities by QPS
  - QPS $= \frac{2}{T}\sum_t (\hat{p}_t - d_t)^2$ Diebold and Rudebusch (1989): the Brier score of a recession probability $\hat{p}_t$ against the NBER indicator d_t (1 in recession); 0 is perfect, 2 the worst
- Clements and Krolzig (1998): MS models rarely beat linear AR in point forecasts of GDP; their value is in densities and in turning-point probabilities

US GDP: one-step density forecasts, 1990–2019



- Expanding window from 1947Q2, 120 quarters, AR(1) re-estimated each quarter, MSIH(2)-AR(1) every 4 quarters (EM, warm start); left: cumulative log-score difference; right: PIT histograms (dashed: uniform)

ATS_ch7_forecast

Interpreting the forecast comparison

- RMSE: AR(1) 0.554, MSIH(2)-AR(1) 0.558: the point forecasts are almost the same, as Clements and Krolzig found
- Mean log score: -1.028 against -0.890; Diebold–Mariano on the difference 1.16 ($p = 0.24$)
- The gain comes from the variance regime: the AR(1) density is too wide after 1984 (PIT piled in the middle, KS $p < 0.001$); the MS density is closer to uniform but not calibrated either (KS $p = 0.01$)
- Cumulative log-score gain before 2008: 17.72; from 2008: -1.10: the regime model pays in calm times and loses in the crisis

An open question

- Can a regime model date Romanian recessions and inflation regimes in real time better than simple rules, and is the 2025 episode a new regime?
  - formal: H_0: equal QPS of the filtered regime probability and of a two-negative-quarters rule against a reference chronology, on pre-registered quarters
  - falsified by a significant difference on quarters the model never saw, with GDP vintages
- Why it matters: Romania has no official business-cycle dating committee; policy reacts to regimes it can see only with delay
  - literature to start from: Hamilton (1989), Chauvet and Piger (2008), Clements and Krolzig (1998), Hamilton (2016)

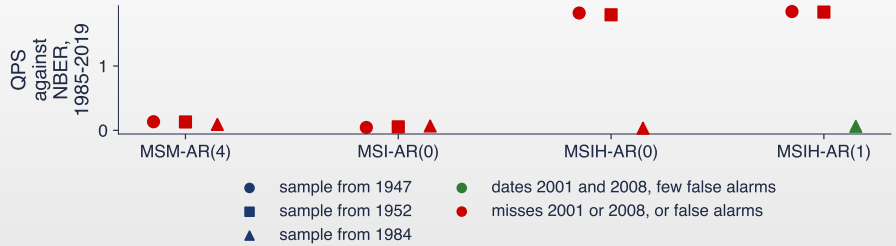
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers that date business cycles of Central and Eastern European countries with Markov-switching models; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Which specification (MSM, MSIH, TVTP with which leading indicator) should separate recessions from volatility eras in Romanian GDP?`
  - **code and replication**: ask for a Hamilton filter, then reproduce a known number first (Hamilton's log-likelihood -181.26)
  - **robustness and critique**: `Act as a hostile referee: list the ways a regime dating could be an artefact of the specification or of the starting values.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The likelihood is the global maximum among many starts, and the chosen regime is not degenerate
- Real-time claims use filtered probabilities and data vintages, never smoothed probabilities
- The number of regimes is tested by bootstrap or chosen by a criterion fixed in advance, not by a χ^2 count
- The meaning of each regime is checked against external events, not assumed from its label

Mini-case: how robust is a recession dating?



- US GDP growth: four specifications × three sample starts, all ending in 2019Q4; QPS of the smoothed low-mean-regime probability against the NBER quarters, 1985–2019
- QPS from 0.034 to 1.846 (a constant probability scores 0.145)
  - 4 variants flag most quarters: their low-mean regime is the Great Moderation
  - 1 of 12 dates both 2001 and 2008 without false alarms: an AI summary that reports one variant as "the" dating is wrong

ATS_ch7_hamilton

Project idea

- **A regime-based business-cycle chronology for Romania**: replicate first, then extend
  - replicate: Hamilton's Table I on his data (log-likelihood -181.26) and the Filardo TVTP model (-586.20)
  - extend: Romanian GDP and industrial production, MSM against MSIH, TVTP with the ESI as leading indicator, Bayesian K selection, real-time filtered probabilities
  - pre-register: sample, specifications, starting-value design, the reference chronology and the QPS comparison
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- The Hamilton filter and the Kim smoother are exact Bayesian recursions for a discrete latent state
- EM and numerical ML find local and degenerate maxima: many starts, then judgement
- The number of regimes is a non-standard testing problem: bootstrap, Garcia, CHP or information criteria
- TVTP, MS-VAR and MS-GARCH make the regimes economic; Gibbs sampling makes the uncertainty visible
- Regimes, breaks and long memory are observationally close; forecasts decide

Self-assessment

Questions

- Why does the Kim smoother need $\hat{\xi}_{t+1|t}$ in the denominator?
- Why does Hamilton's MSM-AR(4) need 32 states?
- Which three conditions of the LR asymptotics fail when testing one regime against two?
- Why is the naive MS-GARCH likelihood path-dependent?
- Why can a regime-switching mean produce a positive GPH estimate?

Next:

- Advanced volatility modelling
- Realised measures, HAR and multivariate GARCH

Appendix: EM for the Markov chain, step by step

- Expected complete-data log-likelihood:
  $Q(\theta \mid \theta^{(m)}) = \sum_j \hat{\xi}_{j,1|T} \ln \rho_j + \sum_{t \geq 2} \sum_{i,j} \hat{\xi}_{ij,t|T} \ln p_{ij} + \sum_t \sum_j \hat{\xi}_{j,t|T} \ln \eta_{jt}$
- Maximise $\sum_{t,j} \hat{\xi}_{ij,t|T} \ln p_{ij}$ subject to $\sum_j p_{ij} = 1$: Lagrangian $\Rightarrow p_{ij} = \sum_t \hat{\xi}_{ij,t|T} / \sum_t \sum_k \hat{\xi}_{ik,t|T}$
- and $\sum_k \hat{\xi}_{ik,t|T} = \hat{\xi}_{i,t-1|T}$; the Gaussian part gives weighted least squares
- Monotonicity: $\ln L(\theta) = Q(\theta \mid \theta^{(m)}) - H(\theta \mid \theta^{(m)})$, and H is maximised at $\theta^{(m)}$ (Gibbs inequality) Dempster, Laird and Rubin (1977)

Back

Appendix: the Kim smoother, the Markov step

- $\Pr(S_t = i \mid S_{t+1} = j, Y_T) = \dfrac{f(y_{t+1:T} \mid S_t = i, S_{t+1} = j, Y_t)\Pr(S_t = i \mid S_{t+1} = j, Y_t)}{f(y_{t+1:T} \mid S_{t+1} = j, Y_t)}$
- If $y_{t+1:T}$ depends on S_t only through S_{t+1} (true on the expanded state), the density ratio is 1
- Then $\Pr(S_t = i \mid S_{t+1} = j, Y_t) = \hat{\xi}_{i,t|t} p_{ij} / \hat{\xi}_{j,t+1|t}$; multiply by $\hat{\xi}_{j,t+1|T}$ and sum over j
- With a continuous state (MS state space, Chapter 6) the ratio is not 1: Kim's smoother is then an approximation

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